

Understanding Offensive Success in NCAA Lacrosse Through Possession-Level Analysis

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1 Introduction to the Data

1.1 Previous Works

Lacrosse is a fast sport, and small details within a possession can have a big effect on whether a team scores. Because of that, it makes sense to study offense at the possession level instead of only looking at final box score stats like goals or shooting percentage. Earlier lacrosse analytics work has shown that shot-based metrics can add useful information that basic box score numbers miss. For example, Burns, Coughlin, and Bolte (2021) found that expected goals style measures gave more information about attacking performance than simple shot totals. In a separate shot quality study, Schanzenbach et al. (2017) also argued that lacrosse teams can learn useful things from studying where shots come from, how shots are created, and how likely different chances are to become goals.

Those ideas helped shape our project. We were interested in the question: what aspects of a possession make a team more likely to score? This question matters because a coach does not only care about whether a team scored; a coach also wants to know what happened during the possession to lead to the result. Lacrosse Reference (2019) noted that shots per possession seems to be tied to other important offensive ideas such as shooting percentage, turnover rate, time of possession, and win rate. That suggests offense is not just about taking more shots or having the ball longer; it is about how a possession is used and what kind of chance it creates.

Our project builds on that idea by using our own observed possession data instead of relying only on team season totals or box score summaries. We wanted to look inside the possession and see whether certain features seemed to lead to stronger outcomes. More specifically, we wanted to study whether possession features were related to whether the possession ended in a turnover, a shot off goal, a save, or a goal. We hoped this would help us find patterns that could actually be useful to a coach trying to improve offensive decision-making and create better scoring chances.

1.2 Our Goal

The goal of our project was to study what happens during a lacrosse possession and see which parts of that possession are connected to better offensive results. Our main research question was: What aspects of a possession make a team more likely to score? We were interested in this because not all possessions are used the same way. Some end quickly, some involve more passing, and some create stronger scoring chances than others. We wanted to see whether those differences showed up in our data.

To study this, we looked at possessions and grouped the outcomes into four categories: Turnover, shot off goal, save, and goal. This let us compare possessions that ended badly with possessions that at least created a shot, and also compare shot-ending possessions that became

goals with ones that did not. In other words, we were not only interested in whether a team scored. We also wanted to understand what kind of possession led to that result.

Before analyzing the data, we expected that some features of a possession would be related to offensive success. In particular, we thought that variables such as possession start type, time in transition, number of passes in transition, time on offense, number of passes on offense, and total possession time might be related to whether a possession ended in a turnover or a shot. We also expected that, among possessions that ended in a shot, variables such as shot type and shot location might help explain whether that shot became a goal, was saved, or missed the goal.

Based on that, our first null hypothesis was that possession features such as possession start type, time in transition, passes in transition, time on offense, passes on offense, and total possession time have no relationship with whether a possession ends in a shot. Our first alternative hypothesis was that at least some of these possession features are related to whether a possession ends in a shot. Our second null hypothesis was that shot type and shot location have no relationship with shot outcome, while our second alternative hypothesis was that shot type and shot location do have a relationship with shot outcome.

1.3 Data Preview

Game Date	Attacking Team	Defending Team	Diff.	Quarter	Pl. Diff.	Pl. Diff. Tm.	Possession Start	Trans. Tm.	Tm. Off.	Poss. Tm.	Trn. Pass	P.A.T.	Poss. Otcn.
4/4/2026	North Carolina	Syracuse	1	1st	0	0	Faceoff	1	47	48	0	9	Shot
05/26/2025	Cornell	Maryland	0	2nd	0	0:00	Settled Offense	0	10	10	0	1	Shot
05/27/2023	Duke	Penn State	0	1st	0	0:00	Faceoff	2	12	14	0	1	Shot
04/04/2026	Notre Dame	Richmond	8	2nd	0	0:00	Faceoff	0	42	42	0	5	Turnover
3/21/2026	Virginia	Utah	0	1st	0	0:00	Turnover	3	8	11	0	1	Shot

Sht Loc	Shot Type	Shot Outcome	# Dodges	Slide Forced	Times thru X	Player Pos.	Player Shts	Player Goals	Player SoG	Player Sht%	Player SoG%
Top Center	Outside Shot	Save	0	FALSE	1	Attack	93	33	55	0.35	0.59
Off Frame	Assited	Missed (Ret. Poss.)	1	FALSE	0	Attack	78	31	58	0.40	0.74
Off Frame	Off Dodge	Missed (Lost Poss.)	1	TRUE	0	Midfield	109	36	74	0.33	0.68
N/A	N/A	N/A	2	FALSE	1	N/A	N/A	N/A	N/A	N/A	N/A
Bot. Center	Off Dodge	Made	1	TRUE	0	Attack	75	32	48	0.43	0.64

Table 1 - A 26 column, 5 row table showing our variables and typical responses.

Our data featured 26 total variables; 3 indexing variables, including Date, Attacking Team, and Defending Team, 5 historical variables, including Player Position, Player Season Shots, Player Season Goals, Player Season Shots on Goal, Player Shooting %, and Player Shots on Goal %. The remaining 18 were observed variables from the games we watched, which included Score Differential (Diff.) Quarter, Player Differential (Pl. Diff., whether there was a power play), Player differential Time (Pl. Diff. Tm., time on said power play), Possession Start (Fashion in which the possession started, included Fast Break, Turnover, Settled Offense, Faceoff, Save, Penalty), Time in Transition (Trans. Tm., seconds), Time in Offense (Tm. Off, seconds), Total Possession Time (Poss. Tm., seconds), Passes in Transition (Trn. Pass), Total Passes after Touch (P.A.T.), Possession Outcome (Poss. Otcn., with options being Shot & Turnover), Shot Location (Sht. Loc, included the 9 quadrants, like upper left, etc. and off frame), Shot Type (Transition, Outside, Assisted, Crease, Off Dodge), Shot Outcome (Goal, Save,

Missed & Retained, Missed & Lost), # of Dodges, whether a Slide was Forced (T/F), and the # of times the ball was Passed through X, the spot behind the goal (Times thru X).

1.4 Data Collection

We collected our data from March 25 to April 8 by watching full NCAA lacrosse game footage from post-COVID seasons on YouTube and ESPN+. Our group studied the game at the possession level, so each row in our dataset represents one offensive possession. In total, we collected 205 observations. Each group member was responsible for collecting 50 possessions, so the observation work was split approximately evenly across the four members of the group.

To gather the data, each group member watched game film and recorded the same set of variables in a shared spreadsheet. We grouped each possession into one of four final outcomes: Turnover, shot off-goal, save, or goal. Since we collected the data ourselves by watching film, many of our variables describe details of the possession that would not normally appear in a basic box score.

Several of our variables were collected directly through observation. The attacking team was the team with possession of the ball, and the defending team was the team without possession. Score differential was (score of the team with possession) - (score of the team without possession). Player differential was calculated as (number of active players on the team with possession) - (number of active players on the team without possession), and player differential time was the time spent during the possession with these conditions. Possession start type described how the possession began, such as after a missed shot, save, turnover, penalty, etc. Time in transition measured how long the team with possession spent on the defensive half of the field before getting the ball to its offensive side. Time on offense measured how long the team with possession spent on its offensive side of the field while still in control of the ball. We also recorded both the passes in transition and the passes on offense. Total possession time was measured as (time in transition) + (time on offense). We recorded the possession outcome as either a shot or a turnover. For possessions that ended in a shot, we recorded shot type, which described the circumstances leading up to the shot, such as whether it came after a dodge, whether it was assisted, whether it was taken from far away, or whether it was taken close to the crease. We also recorded shot location, which described where the shot was aimed if it was on goal, such as top left, top right, middle left, center, etc.

In addition to the variables we observed directly, we also added some player season statistics to the dataset for the player who took the final shot on possessions that ended in a shot. These added variables were player shots, player goals, player shots on goal, and player shooting percentage for the season in which the observed game was played. We looked up these statistics on the teams' official websites. These were not observed from the film itself, so we treated them differently from the possession-level variables that were collected by watching the games. Separating the observed possession data from the added player statistics helped us keep clear which variables came directly from our own data collection and which were included later for extra context.

2 Summary of the Data

2.1 Summary Tables

NUMERICAL		Mean													
Variable	Diff	Pl. Diff.	Pl. Diff. Tm.	Trans. Tm.	Tm. Off.	Poss. Tm.	Tm. Pass.	P.A.T.	# Dodges	Times Thru X	Player Shts*	Player Gals*	Player SoG*	Player Sht%*	SoG %*
Goals	0.71	0.00	1.76	3.71	29.55	30.96	0.63	5.92	1.41	0.80	60.59	23.14	40.14	0.41	0.68
Saves	-0.21	-0.02	1.91	3.81	31.77	33.83	0.98	6.55	2.00	0.74	67.02	26.48	43.87	0.51	0.73
Missed Shots	-0.24	0.02	0.61	4.47	32.73	39.02	1.00	5.96	2.24	0.73	58.92	22.35	38.92	0.38	0.65
Turnovers	0.67	-0.03	1.55	5.24	30.31	31.05	1.47	4.45	1.24	0.83	x	x	x	x	x
Overall	0.26	-0.01	1.46	4.35	31.03	33.57	1.03	5.67	1.70	0.78	62.16	23.88	40.85	0.43	0.68

Table 2 - Mean of all numeric variables by outcome.

NUMERICAL		SD													
Variable	Diff	Pl. Diff.	Pl. Diff. Tm.	Trans. Tm.	Tm. Off.	Poss. Tm.	Tm. Pass.	P.A.T.	# Dodges	Times Thru X	Player Shts*	Player Gals*	Player SoG*	Player Sht%*	SoG %*
Goals	3.64	0.20	9.32	5.50	15.98	17.32	1.22	3.87	1.15	0.85	29.57	15.09	21.17	0.41	0.24
Saves	3.74	0.25	7.41	4.72	16.61	17.86	1.50	4.63	1.63	0.99	41.82	17.62	25.32	0.70	0.35
Missed Shots	3.24	0.14	4.29	5.38	17.16	18.18	1.41	3.93	1.75	0.78	37.01	17.66	25.47	0.43	0.27
Turnovers	3.35	0.18	8.75	6.15	24.27	20.65	1.74	3.64	1.65	0.90	x	x	x	x	x
Overall	3.49	0.20	7.72	5.50	18.98	18.79	1.51	4.05	1.60	0.88	36.09	16.70	23.79	0.52	0.29

Table 3 - Standard Deviation of all numeric variables by outcome.

NUMERICAL		Min													
Variable	Diff	Pl. Diff.	Pl. Diff. Tm.	Trans. Tm.	Tm. Off.	Poss. Tm.	Tm. Pass.	P.A.T.	# Dodges	Times Thru X	Player Shts*	Player Gals*	Player SoG*	Player Sht%*	SoG %*
Goals	-5	-1	0	0	1	1	0	0	0	0	4	2	3	0.2340425532	0.530612
Saves	-13	-1	0	0	3	3	0	0	0	0	2	0	1	0	0.5
Missed Shots	-8	0	0	0	2	2	0	0	0	0	1	0	0	0	0
Turnovers	-8	-1	0	0	0	1	0	0	0	0	x	x	x	x	x
Overall	-13	-1	0	0	0	1	0	0	0	0	1	0	0	0	0

Table 4 - Minimum of all numeric variables by outcome.

NUMERICAL		Max													
Variable	Diff	Pl. Diff.	Pl. Diff. Tm.	Trans. Tm.	Tm. Off.	Poss. Tm.	Tm. Pass.	P.A.T.	# Dodges	Times Thru X	Player Shts*	Player Gals*	Player SoG*	Player Sht%*	SoG %*
Goals	12	1	60	20	57	75	5	15	5	3	129	78	100	3.25	2.25
Saves	8	1	30	17	63	75	6	24	7	5	190	82	116	3.5	2.5
Missed Shots	8	1	30	22	73	78	5	18	8	2	190	82	116	3.25	2.25
Turnovers	11	0	60	24	85	104	6	15	6	3	x	x	x	x	x
Overall	12	1	60	24	85	104	6	24	8	5	190	82	116	3.5	2.5

Table 5 - Maximum of all numeric variables by outcome.

Quarter Count					Quarter %				
Variable	1st	2nd	3rd	4th	Variable	1st	2nd	3rd	4th
Goals	15	11	10	15	Goals	0.29	0.22	0.20	0.29
Saves	13	14	11	9	Saves	0.28	0.30	0.23	0.19
Missed Shots	10	15	12	12	Miss. Shts.	0.20	0.31	0.24	0.24
Turnovers	14	14	17	13	Turnovers	0.24	0.24	0.29	0.22
Overall	52	54	50	49	Overall	0.25	0.26	0.24	0.24

Tables 6 + 7- Count of each quarter by outcome + proportion of each quarter by outcome.

Poss Start Count							Poss Start %						
Variable	Fast Bre.	Turnover	Settled Off.	Faceoff	Save	Penalty	Variable	Fast Break	Turnover	Settled Off.	Faceoff	Save	Penalty
Goals	1	18	7	16	6	3	Goals	0.02	0.35	0.14	0.31	0.12	0.06
Saves	0	12	14	11	6	4	Saves	0.00	0.26	0.30	0.23	0.13	0.09
Missed Shots	0	14	8	19	7	1	Miss. Shts	0.00	0.29	0.16	0.39	0.14	0.02
Turnovers	0	23	4	20	11	0	Turnovers	0.00	0.40	0.07	0.34	0.19	0.00
Overall	1	67	33	66	30	8	Overall	0.00	0.33	0.16	0.32	0.15	0.04

Tables 8 + 9 - Count of each possession shot type by outcome + proportion of each possession by outcome.

Shot Location Count (N=147)											
Variable	TL	TC	TR	ML	MC	MR	BL	BC	BR	Off Target	
Goals	4	4	13	5	1	6	8	5	5	0	
Saves	4	5	3	4	7	6	5	8	5	0	
Missed Shots	1	0	2	3	0	0	1	0	0	42	
Overall	9	9	18	12	8	12	14	13	10	42	

Table 10 - Count of each shot location by outcome.

Shot Location % (N=147)											
Variable	TL	TC	TR	ML	MC	MR	BL	BC	BR	Off Target	
Goals	0.08	0.08	0.25	0.10	0.02	0.12	0.16	0.10	0.10	0.00	
Saves	0.09	0.11	0.06	0.09	0.15	0.13	0.11	0.17	0.11	0.00	
Missed Shots	0.02	0.00	0.04	0.06	0.00	0.00	0.02	0.00	0.00	0.86	
Overall	0.06	0.06	0.12	0.08	0.05	0.08	0.10	0.09	0.07	0.29	

Table 11 - proportion of each shot location by outcome.

Shot Type Count (N=147)						Shot Type % (N=147)					
Variable	Transition	Assisted	Outside Shot	Crease Shot	Off Dodge	Variable	Transition	Assisted	Outside Shot	Crease Shot	Off Dodge
Goals	2	11	15	6	17	Goals	0.04	0.22	0.29	0.12	0.33
Saves	0	10	16	3	18	Saves	0.00	0.21	0.34	0.06	0.38
Missed Shots	1	8	11	2	27	Missed Sh	0.02	0.16	0.22	0.04	0.55
Overall	3	29	42	11	62	Overall	0.02	0.20	0.29	0.07	0.42

Tables 12 + 13 - Count of shot type by outcome + proportion of shot type by outcome.

Slide Forced Count			Slide Forced %		
Variable	TRUE	FALSE	Variable	TRUE	FALSE
Goals	20	31	Goals	0.39	0.61
Saves	15	32	Saves	0.32	0.68
Missed Shots	26	23	Missed Sho	0.53	0.47
Turnovers	0	58	Turnovers	0.00	1.00
Overall	61	144	Overall	0.30	0.70

Tables 14 + 15 - Count of slide forced boolean by outcome + proportion of slide forced boolean by outcome.

Player Position* Count (N=147)					Player Position* % (N=147)				
Variable	Attack	Atk/MF	Midfield	Defense	Variable	Attack	Attack/Mid	Midfield	Defense
Goals	27	3	21	0	Goals	0.53	0.06	0.41	0.00
Saves	29	2	16	0	Saves	0.62	0.04	0.34	0.00
Missed Shots	25	0	21	3	Missed Sho	0.51	0.00	0.43	0.06
Overall	81	5	58	3	Overall	0.55	0.03	0.39	0.02

Tables 16 + 17 - Count of player position by outcome + proportion of player position by outcome.

The summary tables show that our dataset was fairly balanced across quarters, with 52 possessions in the first quarter, 54 in the second, 50 in the third, and 49 in the fourth. In total, the dataset included 205 possessions. Of these, 147 ended in a shot and 58 ended in a turnover. Within the shot-ending possessions, the outcomes were also fairly balanced, with 51 goals, 47 saves, and 49 missed shots. This gave us a good mix of both successful and unsuccessful offensive possessions to compare.

The possession start table also shows where most of our observations came from. The most common possession start types were turnovers and faceoffs, with 67 and 66 possessions,

while settled offense accounted for 33 possessions and saves accounted for 30. Penalty possessions were much less common, and fast breaks were very rare in our sample. This matters because it shows that our dataset mostly reflects more common game situations instead of unusual cases.

Looking only at the 147 possessions that ended in a shot, the shot location table shows that off-frame shots were the single most common shot-location category, making up 42 of the 147 shots, or about 29%. Among shots that were on frame, top right was the most common location, followed by bottom left and bottom center. The shot type table shows that off-dodge shots were the most common shot type by far, with 62 of the 147 shots, or about 42%, followed by outside shots at 29% and assisted shots at 20%. Crease shots and transition shots were much less common. These tables suggest that many shot-ending possessions were created through individual attacking actions, especially dodges, rather than through transition or crease looks.

The numeric summary table helps show what a typical possession looked like. Across outcomes, possessions usually involved around 4 seconds in transition, around 31 seconds on offense, and about 34 seconds total. The average possession also had about 1 pass in transition, about 6 passes after the offense settled, and about 2 dodges or fewer. When the outcomes are compared, some differences start to stand out. Goal, save, and missed-shot possessions all had more passes after touch and more dodges on average than turnover possessions, while turnovers had more passes in transition. These patterns helped guide the later analysis because they suggested that the way a possession developed once the offense got settled may have mattered more than simple possession length alone.

2.2 Figure 1: Possession Features and Whether a Possession Ended in a Shot

To decide which variables to include in Figure 1, we first used R to compare the main possession features between possessions that ended in a shot and possessions that ended in a turnover. For the numeric variables, we used Welch two-sample t-tests in R. We used Welch t-tests because they compare the means of two groups without requiring the groups to have equal variances. For the categorical variable possession start type, we used a chi-square test of independence in R, since that variable is not numeric and we wanted to test whether the proportion of possessions ending in a shot changed across different start types.

Based on those comparisons, we chose to include passes after touch, number of dodges, passes in transition, and possession start type in Figure 1. We chose passes after touch because it showed the strongest numeric difference between the two groups. Possessions ending in a shot averaged 6.17 passes after touch, compared with 4.40 for turnovers, and this difference was statistically significant ($p = 0.0028$). We chose the number of dodges because it also showed a significant difference, with shot-ending possessions averaging 1.88 dodges compared with 1.24 for turnovers ($p = 0.0130$). We included passes in transition because it was also significant and showed the opposite pattern. Turnovers averaged 1.47 passes in transition, while shot-ending

possessions averaged 0.86 ($p = 0.0206$). Finally, we included possession start type because even though it was not statistically significant at the 0.05 level, it came close and still showed a noticeable pattern in the data ($\chi^2(5) = 10.18$, $p = 0.0704$).

We left out time in transition, time on offense, and total possession time because those variables did not show strong enough differences between shot-ending possessions and turnovers to justify featuring them in Figure 1. Their p-values were 0.1753, 0.7716, and 0.2585, respectively. Since our goal for Figure 1 was to highlight the possession features that gave the clearest evidence of a relationship with whether a possession ended in a shot, we focused on the variables with the strongest statistical support and the clearest visual separation between the two groups.

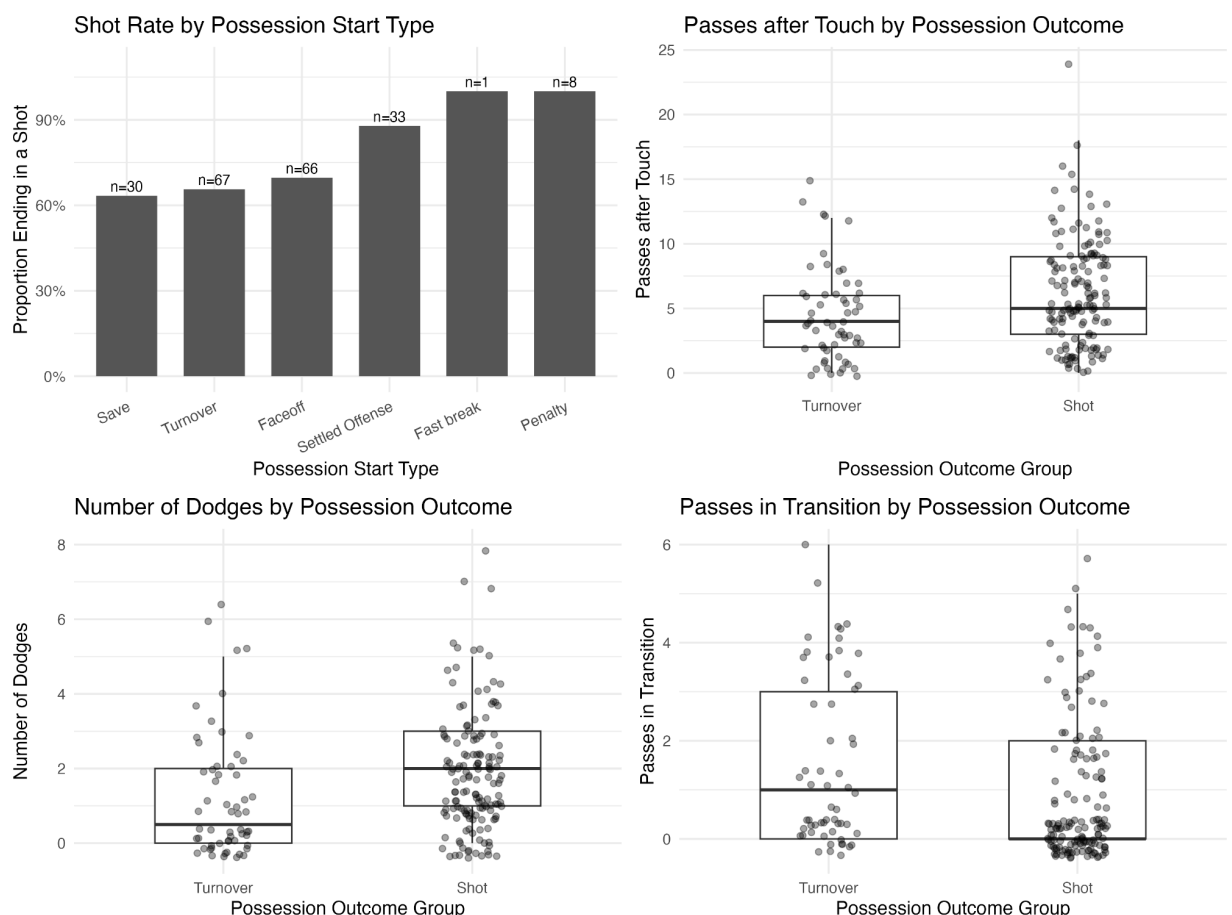


Figure 1. Four-panel figure showing the proportion of possessions ending in a shot for each possession start type in the plot on the top left, as well as the distribution of passes after touch, number of dodges, and passes in transition for possessions ending in a shot or turnover in the top right, bottom left, and bottom right plots, respectively.

Figure 1 shows these relationships in four panels. The first panel shows the proportion of possessions ending in a shot for each possession start type. Among the more common start types, settled offense had the highest shot rate, while possessions beginning after a save had the lowest.

Fast break and penalty possessions had very high shot rates, but those categories had very small sample sizes, so those results should be interpreted carefully. The other three panels compare the numeric variables between possessions ending in a shot and possessions ending in a turnover. In the passes after touch panel, shot-ending possessions have a noticeably higher center and a wider upper range than turnovers. In the number of dodges panel, shot-ending possessions again appear to have a higher center, showing that possessions with more dodging were more likely to end in a shot. In contrast, the passes in transition panel shows that turnovers tended to involve more transition passing, while many shot-ending possessions had very few passes in transition.

Overall, Figure 1 suggests that the actions that happened after the offense got settled were more important than simple possession length for determining whether a possession ended in a shot. More passing after the offense was established and more dodging were both associated with shot creation, while more passing in transition was associated with turnovers. This made these variables the clearest choices to feature in Figure 1 and to discuss further in the next section.

2.3 Figure 2: Shot features and Whether a Shot ended with a Goal

After investigating what possession features lead to a shot, we wanted to investigate what makes a shot more likely to lead to a goal. Shot features that we used included shot location, which divided the goal into 9, 2'x2' squares as well as missed goals; full options for this variable are cataloged in Table 10. The other shot feature was the type of shot taken, with possible values of transition, crease, outside, off dodge, or assisted.

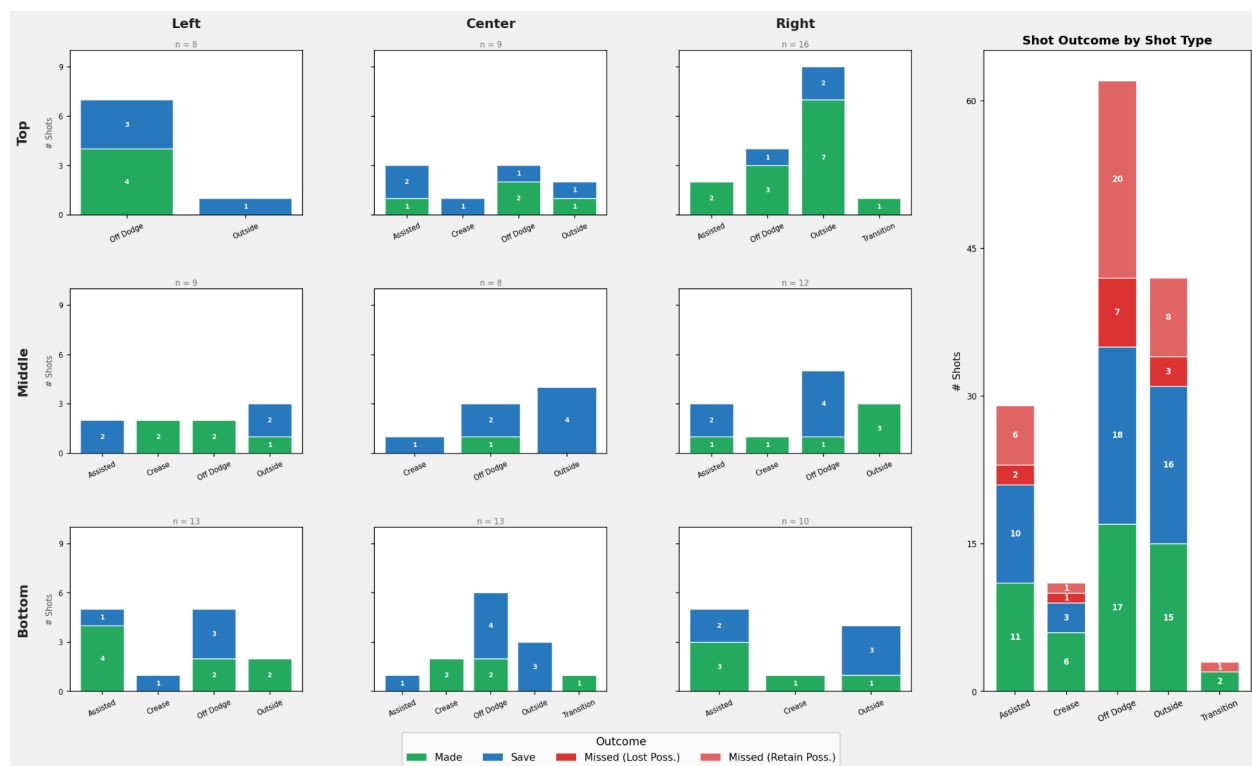


Figure 2. Stacked bar chart showing made, saved, missed and lost possession, and missed and retained possession (green, blue, dark red, and light red, respectively) in each section of the goal, as well as the shot outcome for each shot type in any location in the plot on the right.

In the location grid, the top-right stands out as the region with the highest proportion of made shots, 81.25% on 16 shots, while the middle center is dominated by saves, with a make percentage of just 12.5% on 8 shots. In the shot type panel, it can be seen that crease shots have the highest make percentage with 54.5%, but only 11 shots, far fewer attempts than assisted, off dodge, and outside shots that had 29, 62, and 42 shots respectively. These three types have a more even split of made, missed and saved. For example, of all outside shots, 15 (35.7%) were made, 16 (38.1%) were saved, and 11 (26.2%) were missed. Transition shots accounted for only 3 of 147 shot attempts recorded.

3 Insights from the Data

3.1 Insight 1: Which Possession Features Made a Possession More Likely to End in a Shot?

One of the clearest things we learned from our data was that possessions ending in a shot looked different from possessions ending in a turnover, especially after the offense got settled. As shown in Figure 1 and the summary statistics in Table 2 and Table 3, the biggest differences were in passes after touch, number of dodges, and passes in transition. To test these differences, we used Welch two-sample t-tests in R for the numeric variables and a chi-square test of independence in R for possession start type.

The strongest result was for passes after touch. Possessions ending in a shot averaged 6.17 passes after touch, while possessions ending in a turnover averaged 4.40. This difference was statistically significant, with $p = 0.0028$. We also found that possessions ending in a shot had significantly more dodges, with an average of 1.88 compared with 1.24 for turnovers ($p = 0.0130$). In contrast, possessions ending in a turnover had more passes in transition on average than possessions ending in a shot, with means of 1.47 and 0.86, respectively ($p = 0.0206$). The time-based variables did not show strong evidence of a difference. Time in transition had $p = 0.1753$, time on offense had $p = 0.7716$, and total possession time had $p = 0.2585$. This suggests that what the offense did during the possession mattered more than how long the possession lasted.

We also looked at possession start type to see whether the way a possession began was related to whether it ended in a shot. The chi-square test gave $\chi^2(5) = 10.18$ with $p = 0.0704$. This was not statistically significant at the 0.05 level, but it was close enough to suggest that there may still be a meaningful pattern. In Figure 1, possessions beginning in settled offense had one of the highest shot rates among the more common categories, while possessions beginning after a save had the lowest. At the same time, fast break and penalty possessions had very high shot rates, but those groups had very small sample sizes, so we do not want to overinterpret them.

To look at the main numeric variables together, we also fit a logistic regression model in R with shot-ending possession as the response variable and passes after touch, number of dodges, and passes in transition as predictors. This model supported the same general pattern. Passes after touch remained significant ($p = 0.025$), and passes in transition also remained significant ($p = 0.006$), while number of dodges was no longer significant after controlling for the other two variables ($p = 0.081$). The model estimated that each additional pass after touch increased the odds of a possession ending in a shot by about 11.6% (odds ratio = 1.12, 95% CI: 1.01 to 1.23). On the other hand, each additional pass in transition decreased the odds of a possession ending in a shot by about 24.8% (odds ratio = 0.75, 95% CI: 0.61 to 0.92).

In plain lacrosse terms, this suggests that possessions were more likely to create a shot when the offense was able to do more after it got into the offensive area. More passing after the offense was set and more dodging were both associated with possessions ending in a shot, while more passing in transition was associated with turnovers. From a coaching point of view, this could mean that shot creation depends more on effective half-field offense than on simply moving the ball around in transition or extending the possession. In our data, the strongest possessions were not necessarily the longest ones. They were the ones that showed more purposeful offensive action once the team got settled.

3.2 Insight 2: Which Shot Features Made a Possession More Likely to End in a Goal?

The most important insight from the shot data is that shot location appears to be more influential than shot type when it comes to whether or not that shot produces a goal. In figure 2, as mentioned before, the top right region has the highest made-shot percentage, while the middle center region was dominated by saves. In comparison, as seen in the right panel of figure 2, for most shot types, the distribution of made, saved, and missed shots were much more even. To further investigate this insight we used chi-square tests of independence for shot location versus shot outcome and shot type vs shot outcome in Python. Additionally we fit a logistic regression model with shot outcome as the response and shot type and shot location as predictors.

The shot type vs shot outcome chi-square test yielded a p-value of 0.6817, meaning that in this sample there is not statistically significant evidence that the shot type and outcome are associated. The p-value of shot location vs shot outcome was $4.8e-17$, when off frame shots were included. This makes sense because the category of “off frame” is also a perfect predictor of whether or not the shot went in, thus for the rest of the analysis we will remove off frame shots from the shot locations. After removing the off frame shots, the p-value of shot location vs shot outcome becomes 0.2497 which is also not statistically significant.

For the logistic regression the category of transition goals was also removed, accounting for only 3 of 147 shots; this sample size was too small and prevented the model from converging. After removing transition goals, the model converged and showed that shot location was the only statistically significant predictor of a made shot. In particular shots taken in the top-right section

of the goal had much higher odds of being made than those taken in the reference location (bottom-center) with a p-value of 0.032 and an odds ratio of 6.3. None of the shot type categories were statistically significant, suggesting that placement mattered more than shot type in this sample.

3.3 Statistical Support / Methods

To support our insights, we used a variety of statistical tests that matched the type of variables and research questions. For the first insight which focused on whether a possession ended in a shot or turnover, we used Welch two-sample t-tests for numerical variables such as passes after touch, number of dodges, and passes in transition. Welch t-tests were used because they allow for differences in variance between groups and are appropriate for comparing means between two independent groups. For categorical variables such as possession start type, we used a chi-square test of independence. This allowed us to determine whether the distribution of possession outcomes differed across categories of how the possession began. In addition to these individual tests, we used a logistic regression model to analyze the combined effect of multiple variables on whether a possession ended in a shot. Logistic regression was appropriate because response variable shot vs turnover was binary. This model allowed us to estimate how each variable affected the odds of a possession ending in a shot while controlling for the other variables.

For insight 2 which focused on whether a shot resulted in a goal, we used chi-square tests to examine relationships between categorical variables like shot type and shot location with shot outcome. We used logistic regression with goal vs non goal as the response variable. This allowed us to determine which shot characteristics were most strongly associated with scoring while accounting for multiple variables at once.

3.4 Critiques / What We Would Do Differently

While our analysis provides useful insights, there are limitations that should be acknowledged. For starters, our dataset consisted of 205 possessions collected manually from a limited number of games. Although this is enough to identify some patterns, a larger dataset across more teams and games would improve the generalizability of our results.

Second, many of our variables were based on manual observation which introduced potential subjectivity and measurement error. For example, variables such as shot type, shot location, and whether a slide was forced depend on the observer's judgement and may not be consistent across different group members.

Third, our data set does not fully capture defensive pressure. While we included some indicators like slides forced and player differential, we did not directly measure how tightly a player was defended or how quickly the defense rotated. These factors may have a major impact on shot quality and turnover probability.

Additionally, some categories had very small sample sizes such as transition shots and fast break possessions. This limited our ability to draw strong conclusions about these situations and made statistical tests less reliable.

If we were to improve this study, we would collect a larger and more balanced dataset, include more detailed measures of defensive pressure, and standardize the way variables are recorded to minimize subjectivity. We would also consider tracking additional factors such as game situation, player matchups, and shot clock timing which could provide deeper insight into offensive decision making.